

STROKE PREDICTION

Stroke Prediction Using Machine Learning

Dataset: Healthcare Stroke Prediction (5109 records)

Models: Naive Bayes, Decision Tree (H2O), Decision Tree (rpart), Logistic Regression

Method: 70:30 train vs test set split on the data + SMOTE

Andile Ndwandwe (217012434)

Sanele Mbuli (216015915)



EDGEWOOD CAMPUS

HOWARD COLLEGE CAMPUS

NELSON R MANDELA SCHOOL OF MEDICINE

PIETERMARITZBURG CAMPUS

WESTVILLE CAMPUS

Problem : Predicting Stroke risk Using Patient Data

- Stroke is a major cause of death and disability worldwide
- It occurs when blood flow to the brain is interrupted (blockage or bleeding)
- Early identification of high –risk patients is critical for prevention

Key risk factors include:

- Age, Hypertension, Heart disease, Glucose levels and lifestyle (e.g. smoking)

Objective

- To develop and evaluate a binary classification model that predicts where a patient will experience stroke (Yes/No) based on medical and demographic factors.

Data Checks

- Missing values
- Encoding requirements
- Variable relevance
- Outliers

Cleaning Steps

- Dropped ID variable (not useful for prediction)
- Handled missing BMI values using median imputation
- Categorical variables were encoded using one hot encoding
- Removed rare gender variable “Other”
- Dropped redundant age group variable in final model
- Dropped work type variable (low predictive relevance)

Dataset

- 5109 records/Observations
- Target Variable : Stroke
 - 1 = Stroke
 - 0= No Stroke

Target distribution

Stroke – 4.87%

No Stroke – 95.13%

Features Used

- Age, Hypertension, Heart disease, Glucose level, BMI, Smoking Status, Residence type, Marital status

Problem : Severe class imbalance

- Stroke cases are rare 4.87%
- Majority class dominates 95.13%
- Model tends to bias toward non stroke predictions

Resampling Methods Evaluated: Using Logistic Regression as a baseline

- Under-sampling
- Over-sampling
- Combined (Over and under) sampling
- SMOTE – Chosen

HANDLING IMBALANCE

Why SMOTE?

Selected based on best AUC performance

Class distribution before and after SMOTE

SPLIT	Class 0 (No Stroke)	Class 1 (Stroke)	Total
Original training set	3402 (95.13%)	174 (4.87%)	3,576
After SMOTE	1788 (50.00%)	1788 (50.00%)	3,576
Test set (unchanged)	1458 (95.11%)	75 (4.89%)	1,533

Balancing technique comparison (LR base model)

Technique	Train AUC	Test AUC	Train MCC	Test MCC
Under-sampling	0.8456	0.8261	0.5240	0.2356
Over-sampling	0.8503	0.8321	0.5553	0.2361
Combined (both)	0.8611	0.8344	0.5779	0.2512
SMOTE	0.8535	0.8323	0.5537	0.2427

Model Training

- Model s trained on SMOTE –balanced training set
- Evaluated on unchanged test set

Machine Learning Techniques used (70:30 train/test split)

- Naïve Bayes
- Decision Tree (H20 tuned)
- Decision Tree (rpart)
- Logistic Regression

Hyperparameter tuning

- Max_depth = 13 (H20 Decision Tree)
- Min_rows = 1 (H20 Decision Tree)
- Threshold = 0.5

AUC COMPARISON AND MODEL SELECTION

Why AUC?

- Threshold- independent metric
- Suitable for imbalanced classification

Model	Train AUC	Test AUC	AUC Gap	Diagnosis
Naive Bayes	0.8319	0.8123	0.0196	Good generalisation
Decision Tree (H2O)	0.9886	0.6659	0.3223	Severe overfitting – high variance
Decision Tree (rpart)	0.7685	0.7346	0.0339	Moderate gap
Logistic Regression	0.8535	0.8323	0.0212	Best bias-variance trade-off

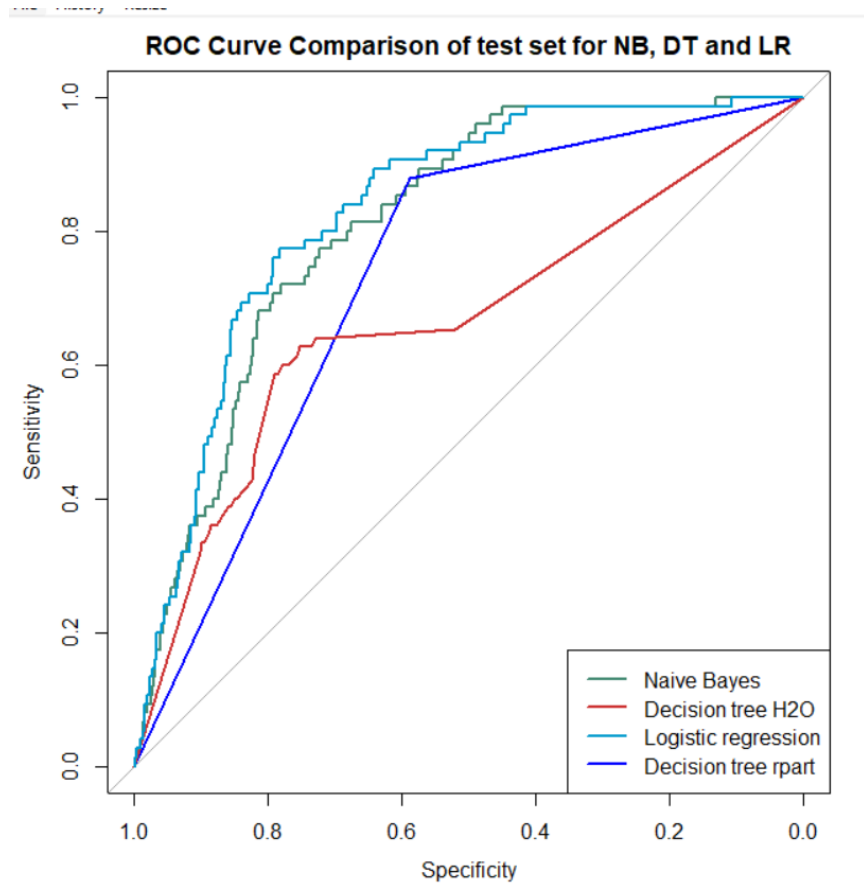
Best Model – Logistic Regression

Model Performance Insights

- All models achieved high recall on the test set
- Precision is low due to highly imbalanced dataset
- Decision Tree H2O shows overfitting
- Logistic Regression performs mostly consistently and has best AUC

Model	Set	Accuracy	Precision	Recall	F1	MCC	AUC
Naive Bayes	Train	0.7576	0.7254	0.8289	0.7737	0.5204	0.8319
	Test	0.6888	0.1135	0.7867	0.1983	0.2144	0.8123
Decision Tree (H2O)	Train	0.9365	0.9363	0.9368	0.9365	0.8730	0.9886
	Test	0.8003	0.1092	0.4267	0.1739	0.1347	0.6697
Decision Tree (rpart)	Train	0.7685	0.7030	0.9295	0.8006	0.5671	0.7685
	Test	0.6034	0.0992	0.8800	0.1784	0.2042	0.7346
Logistic Regression	Train	0.7760	0.7561	0.8149	0.7844	0.5537	0.8535
	Test	0.7306	0.1294	0.7867	0.2222	0.2427	0.8323

CONFUSION METRICS



TOP 5 PREDICTORS

Logistic Regression (Standardised coefficients)

Variable	Std. Coefficient	Odds Ratio	P-value
age	1.7841	1.0838	0.0000
hypertension.1	0.4661	1.5937	0.0001
heart_disease.1	0.3462	1.4137	0.0143
avg_glucose_level	0.2377	1.0043	0.0000
Formerly smoked .1	0.1546	1.1672	0.1497

- Age is the strongest predictor of stroke
- Each additional year increases stroke odds by 8%
- Hypertension is the second strongest predictor
- Formerly smoked $p > 0.05$

- SMOTE was used to address the severe class imbalance 4.9%
- Logistic Regression Performed best, achieving the highest test AUC(0.83) and strong MCC
- Decision Tree showed overfitting with large gap between training and test performance
- Age is the strongest predictor
- Logistic regression with SMOTE provides an interpretable and reliable baseline for stroke predictions,